

basis (see TRSDOS technical information in TRSDOS & Disk BASIC Reference Manual). This facility seems to consistently drop the last sector, so I have found it necessary to pad my files at the end. Check out the LOF function—I haven't tried it on these files.

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BASIC Doodle

In the article "Doodlebug," I felt the unexperienced programmer without machine language orientation was left out. I have developed an 11 second, BASIC, reverse video program for Level II:

```
5 ON ERROR GOTO 120
.
.
.
100 FORB 15360TO16383:POKE B,319 - PEEK(B):NEXT
110 RETURN
120 RESUME NEXT
```

One word of warning: The program will demolish any letters on the screen.

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TC-8 Cassette Interface

I am the owner of an early model TRS-80 microcomputer, which has a CTR-41 cassette recorder instead of a CTR-80.

A few weeks before Christmas I purchased a TC-8 Cassette Interface unit from JPC. I assembled the unit at a friend's home, following the very well written assembly instructions. The system tested flawlessly and just as advertised. I was extremely pleased and delighted with my new hardware and software.

I later connected my TC-8 to my CTR-41 cassette recorder after modifying it in the manner explained in the TC-8 Cassette Operating System users manual. I was surprised to find that the recorder motor control function was not working. The CTR-41 motor ran whenever any of the control buttons were depressed. This malfunction made it impossible to make use of one of the TC-8's nicest features; the ability to position to various program and data files on the tape. After ascertaining that the required modification had been performed correctly, I decided to call the folks at JPC.

I was introduced to a Mr. Bruce Kernodel and we exchanged information to establish exactly what my problem might be. Mr. Kernodel was highly knowledgeable of the inner workings of both the TC-8

and CTR-41. He explained that the CTR-41 modification is designed to connect system ground to power supply ground directly while bypassing the remote jack. He also explained that this leaves only the motor to be switched by the remote jack and thus reduces the amount of current that the TC-8 and TRS-80 must handle in turning the cassette recorder off and on. Mr. Kernodel also mentioned that in the past all of the CTR-41 recorders they had seen were wired exactly the same with regard to the wires connecting the remote jack to the circuit board. He did say, however, that if for some strange reason they were not wired that way in my recorder I would need to reverse them.

I followed Mr. Kernodel's advice and reversed the two wires. I was relieved to discover that the TC-8 now functioned beautifully.

If you have followed all of JPC's instructions correctly and have the problem I have had, locate the two small white wires connected to the remote jack and try reversing them.

My thanks to Mr. Kernodel and the folks at JPC. Their product and support is first rate and deserving of praise.

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Short Stuff

This short utility grew out of a competition between my son and myself, with Jeffrey finally getting the last word when he got it down to 28 characters (including the line number). At that point I thought it just interesting enough that you might want to pass it on to your readers to elaborate on.

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The World's Shortest Wordprocessor:
Written by Jeffrey Vavasour, 11 years old
1 PRINTCHR\$(14);INKEY\$;GOTO 1

The program could be further shortened to remove the cursor, by only typing "PRINT INKEY\$;GOTO 1". Clearing the screen, either in whole or in part, is performed by first positioning the cursor by using SHIFT and one of the arrows, then pressing the CLEAR key which will erase the screen from the cursor position to the bottom of the screen.

Printer output is obtained by using the JKL feature of NEWDOS.

END Your Programs

Your December issue more than paid for the first year's subscription. "Assemble It Yourself" by Richard Koch makes Radio Shack's Editor/Assembler usable for the first time. The article is quite clear, and the Program Listings gave no prob-

lems. Well done!

I also found Richard Halloran's "Printer Pagination" quite useful, and lost no time in using it. No one-upsmanship intended with the enclosed listing of my version of Mr. Halloran's program.

```
10 CLEAR500
20 OPEN"1",1,"PROGRAM"
30 CLS:PRINT@320,"LOADING THE PROGRAM INTO MEMORY
40 PRINT:PRINT"IN CASE YOU FORGOT, FILE NAME IS 'PROGRAM'"
50 PN = PN + 1:LPRINTSTRINGS(70," ") "PAGE
  "PN:LPRINT" "LPRINT" "
60 IFPEEK(16425) = 50 THEN LPRINTCHR$(12)
  :GOTO 50
70 LINEINPUT #1,RS
80 LPRINT RS
90 IF RIGHT$(RS,3) = "END" CLOSE ELSE 60
100 LPRINTCHR$(12)
110 END
```

I would like to offer your readers the following suggestions: Call the last line of the program to be LISTED "END". Then it is not necessary to count (or remember) the number of lines in the program. The printer will stop when the program line is "END" and do a last top-of-form. Line 50 of the new version automatically prints the page number at the top right corner of each page of the listing.

Finally, be sure that the number of characters in each line of your program does not exceed the number of characters per line of the printer, otherwise the pagination will become skewed. My Line Printer II (Centronics 730) will issue a carriage return/linefeed automatically at the end of 80 characters, while the TRS-80 updates location 16425 only when the program calls for a linefeed.

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Two ROM Improved?

The change by Radio Shack from the three ROM version of the TRS-80 to the two ROM improved (?) system blocks reading tapes on the second cassette connected to the interface. They can be written to, but not read. This arbitrary change is contrary to the claims and technical data on the interface, and Radio Shack should be pressured into putting the feature back. In fact, the read function is the most useful aspect of having a second cassette recorder, since it can be used as a software ROM to load utility programs into protected memory. Included in those programs is the printer driver, etc.

Readers should take appropriate steps if they are considering buying an interface with the intention of using all the stated facilities.

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